

解: (1)

DH 参数表

关节	a_{i-1}	$\vec{a}_{i-1}$	d_i	θ_i
1	0	0	0	θ_1
2	0	1	0	θ_2
3	$4\frac{1}{2}^\circ$	0	$\sqrt{2}$	θ_3
4	0	$\sqrt{2}$	0	θ_4

$${}^{i-1}_i T = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ s\theta_i c a_{i-1} & c\theta_i c a_{i-1} & -s a_{i-1} & -s a_{i-1} d_i \\ s\theta_i s a_{i-1} & c\theta_i s a_{i-1} & c a_{i-1} & c a_{i-1} d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0_1 T = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & 0 \\ s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} C_1 & -S_1 & 0 & 0 \\ S_1 & C_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^1_2 T = \begin{bmatrix} C_2 & -S_2 & 0 & 1 \\ S_2 & C_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^2_3 T = \begin{bmatrix} C_3 & -S_3 & 0 & 0 \\ \frac{\sqrt{2}}{2} S_3 & \frac{\sqrt{2}}{2} C_3 & -\frac{\sqrt{2}}{2} & 0 \\ \frac{\sqrt{2}}{2} S_3 & \frac{\sqrt{2}}{2} C_3 & \frac{\sqrt{2}}{2} & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^3_4 T = \begin{bmatrix} C_4 & -S_4 & 0 & \sqrt{2} \\ S_4 & C_4 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0_4 T = \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$p_x = C_1 + S_{12} + \sqrt{2} C_{12} C_3 - S_{12} S_3$$

$$p_y = S_1 - C_{12} + \sqrt{2} S_{12} C_3 + C_{12} S_3$$

$$p_z = S_3 + 1$$

$$r_{11} = C_2 C_{34} - \frac{\sqrt{2}}{2} S_{12} S_{34}$$

$$r_{12} = -C_{12} S_{34} - \frac{\sqrt{2}}{2} S_{12} C_{34}$$

$$r_{13} = \frac{\sqrt{2}}{2} S_{12}$$

$$r_{21} = S_{12} C_{34} + \frac{\sqrt{2}}{2} C_{12} C_{34}$$

$$r_{22} = -S_{12} S_{34} + \frac{\sqrt{2}}{2} C_{12} C_{34}$$

$$r_{23} = -\frac{\sqrt{2}}{2} C_{12}$$

$$r_{31} = \frac{\sqrt{2}}{2} S_{34} \quad r_{32} = \frac{\sqrt{2}}{2} C_{34} \quad r_{33} = \frac{\sqrt{2}}{2}$$

(2) $\hat{R} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$ $R = R_z(\theta_1) R_z(\theta_2) R_x(45^\circ) R_z(\theta_3) R_z(\theta_4)$

$$\theta_3 = \arcsin(p_z - 1) \text{ 或 } \pi - \arcsin(p_z - 1)$$

$$\theta_4 = \arctan 2(r_{31}, r_{32}) - \theta_3$$

$$\theta_1 = \arctan 2 \left[(p_y - \sqrt{2} r_{23} - 2 r_{13} C_3 + \sqrt{2} r_{23} (p_z - 1)), (p_x - \sqrt{2} r_{13} - 2 r_{23} C_3 - \sqrt{2} r_{13} S_3) \right]$$

$$\theta_2 = \arctan 2(\sqrt{2} r_{13}, -\sqrt{2} r_{23}) - \theta_1$$